

9/7/26

① Find out the list of companies which are conducting the placement drive using DSA course.

② Find out the syllabus of Gate for DSA

(1A) List of Companies - Google, Amazon, Microsoft, Adobe, Flipkart, Uber, Oracle, Goldman Sachs, Walmart labs, Samsung R&D, TCS Digital, Infosys (specialist roles) Directi, and Paytm.

(2A) GATE Syllabus for DSA

- Arrays, stacks, queues, linked lists
- Trees - binary trees, BST, balanced trees (AVL, B-trees)
- Graphs: representations, traversals (BFS/DFS)
- Hashing
- Sorting algorithms and their complexities
- Asymptotic notation (Big-O, Omega, Theta)

- Algorithms portion (often paired with DS) also includes divide and conquer, greedy, dynamic programming, algorithm design and analysis.

Q1) Find out set of I/O for the set of following.

1. Electricity Bill
2. Bus ticket
3. Message to a friend

Q2) Today when you started from home to do what are the various datastructures you identified in your surroundings.

Ans

(i) Electricity Bill

Input: previous meter readings, current meter readings, number of units consumed, Tariff rate, Customer ID/acc number, taxes, fixed charges, billing period

Output: Total units consumed, total amount payable, due date, bill number / reference, itemized charges (energy charge + fixed charge + tax)

(ii) Bus Ticket

Input: Source and destination stop, date and time of travel, number of passengers, ticket type (student/general), fare rate per km.

Output: Ticket number, total fare, seat number (if applicable), boarding point, printed/generated ticket.

(iii) Message to a friend

Input: Sender's message/text, recipient's contact number, Sender's network

Output: Message sent confirmation, Delivery status, Time of Sending

2Ans

Data Structures around our surroundings when we started from home to college

Array - Row of parked bikes/cars

Queue - Students waiting at the bus stop or canteen

Tree - Road map with branches

Graph - Road network connecting different places

Hash table - phone contacts stored by name

matrix - Seating arrangement in a classroom

Circular Queue - Round robin traffic at a signal.

Linked List - Chain of buses moving one after another

Stack - Stack of books or plates.